

Can Language Models Simulate Psychometric Respondents?

A Five-Model Sensitivity Study with Baseline and Zero-Prompt Controls

on a Polish Mentalization Battery

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Abstract

Before collecting human data for a new psychometric battery, an author can ask a different question first: do the instruments react to designed personality variation at all? Thirty fictional Polish-language biographies with predetermined profiles across 12 dimensions were processed by five large language models from four vendors — Claude Sonnet 4.6 and Opus 4.6 (Anthropic), GPT-5.4-mini (OpenAI), Grok-4-20-reasoning (xAI), and Gemini 3 Flash (Google) — under three observation conditions: a **persona condition** (each biography presented twice per model; 285 valid runs), a **baseline condition** (each model answered the battery as itself, 7–10 times per model; 46 runs), and a **zero-prompt condition** (the battery presented with no system instruction at all; 31 valid runs). Seven human respondents completed the same battery (exploratory). This is a sensitivity study, not a validation.

The persona condition shows that four published Polish self-report instruments recover the embedded profiles with high consistency (attachment $\kappa = .73$ for four models, $.33$ for GPT, which I discuss as an outlier; median rank-score $r = .71$ to $.82$; literature-consistency 8–10/10). Cross-vendor agreement on questionnaire z -scores is high (median pairwise $r = .78$ – $.96$); cross-vendor agreement on the author-keyed TCTM-22 vignette test is markedly lower ($\text{ICC}(2, 1) = .25$ across all five models, but $.39$ when GPT is excluded as a sensitivity check), with eight of 22 items showing Bonferroni-significant cross-model disagreement. The baseline condition shows that each model has a stable response signature absent any persona (GPT scores anxious-preoccupied in 100% of baseline runs; Gemini hits ceiling on every self-report scale; the two Claude models differ in default attachment style), and that persona instructions expand response variance by a factor of 3.5 to 8.9 over baseline (model-level medians across 9 questionnaire dimensions). The zero-prompt condition shows that the “answer as yourself” instruction itself is associated with systematic shifts in self-report: relative to no instruction, the instruction co-occurs with approximately 0.9 scale points lower attachment avoidance for Sonnet (Glass’s $\Delta \approx 1.96$) and 1.0 scale point higher emotional stability for Gemini (within-condition $\Delta = -1.00$); GPT moves in the opposite direction on mentalization and anxiety. TCTM-22 mean accuracy varies by less than ± 5 percentage points between baseline and zero-prompt for four of five models; GPT shows an 8.1 pp drop without instruction.

Open data, biographies, and analysis scripts are released under CC BY 4.0; every numeric claim in this paper is reproducible from the released CSV.

Keywords: synthetic respondents, instrument sensitivity, mentalization, attachment, cross-vendor LLM comparison, baseline response signature

1 Introduction

Developing a new psychometric instrument requires extensive piloting. A growing line of research suggests that large language models (LLMs) can serve as simulated respondents, providing rapid pre-pilot probes of instrument behavior before committing to human recruitment [1, 3]. The narrow question this paper addresses is not whether LLMs “have” personality, but whether psychometric instruments *respond systematically* to designed personality variation embedded in biographical prompts. If they do, the instruments are at minimum sensitive to the relevant signal — a necessary precondition for construct validity [2]. If they do not, the problem is worth catching before recruiting humans.

Most existing studies use a single model family [1, 3]; I use five models from four vendors (Anthropic, OpenAI, xAI, Google), which lets me ask whether sensitivity findings generalize across architectures or are model-specific artifacts. I also include two control conditions: a **baseline** condition in which the model answers the battery as itself, repeatedly (probing the model’s response signature absent any persona), and a **zero-prompt** condition in which the model receives no system instruction at all (probing what part of the response profile is shaped by even minimal instructional context). Together, the persona / baseline / zero-prompt comparison lets me separate three sources of response variance: inherent within-condition stochasticity, the contribution of instructional framing, and the contribution of biographical prompting.

I apply this design to four published Polish-language instruments — DBZ-R (attachment; Polish ECR-R adaptation), MentS-PL (mentalization), KPP (need for cognition), TIPI-PL (Big Five) — plus one original, author-keyed mentalization vignette test (TCTM-22). Thirty fictional personas are processed twice through each model; seven human respondents complete the same battery as a small exploratory comparison.

2 Method

Personas. 30 biographies in Polish (1,554–2,914 file-total words including YAML metadata; 1,489–2,861 narrative-body words after stripping YAML), written in second person, encoding predetermined profiles across 12 dimensions (11 continuous + 1 categorical): attachment anxiety and avoidance (DBZ-R), mentalization self / other / motivation (MentS-PL), need for cognition (KPP), Big Five E/A/C/ES/O (TIPI-PL), and a four-category attachment style. Biographies were authored by the first author using Claude Opus 4.6 as a writing assistant; the author specified profiles, narrative constraints, and stylistic requirements, and reviewed and edited every draft. Six biographies (adrian, ewa, kuba, magda, pawel, weronika) were revised after pilot review and are tagged `persona_version: v2`; the rest are at v1. The version sent to all five models is the version recorded in the released `all_data.csv`. No biography was designed to contain trait labels; the first author manually flagged nine biographies as containing direct trait-language references (ania, anna-sim, filip, jakub, kamil, marek, natalia, ola, torek; the “explicit” subset) and 21 as “narrative”. A separate broader-keyword regex audit yields a different split; both codings ship in the open dataset.

Instruments. DBZ-R (Polish adaptation of ECR-R; 36 items, Likert 1–7; (author?) 7); MentS-PL (28 items; (author?) 5); KPP (36 items; (author?) 8); TIPI-PL (10 items; (author?) 9). These four use published Polish norms. TCTM-22 is an original instrument: 22 vignettes scored against author-defined answer keys, with each option tagged under the MASC error taxonomy [4]: *DOS* (under-mentalizing,

reduced theory of mind), *NAD* (over-mentalizing, excessive theory of mind), and *BK* (no mentalizing). The 22 vignettes cover five subscales (4–5 items each): subtext recognition, manipulation recognition, waning-bond detection, pursuit–withdrawal dynamics, and authenticity of relational repair. **Throughout this paper, all TCTM-22 results refer to author-keyed performance, not validated mentalization;** cross-vendor disagreement on which items are hard (§3.2) is itself evidence that the keyed answers may not reflect a single shared construct.

Ground truth and STYLE_MAP. Each persona’s expected attachment style is taken from the top-level `expected_style` field in `analysis_summary.jsonl`. Four personas (Ewa, Kamil, Marek, Radek) carry `disorganized` in their YAML `expected_profile` block but `fearful_avoidant` in the top-level `expected_style`; this maps `disorganized` to `fearful-avoidant` under the four-category Bartholomew taxonomy that all five models use in their classifier outputs. Under a strict five-category policy (treating `disorganized` as a fifth class no model can produce), all attachment κ values would drop by approximately 10 percentage points; I report both in the released supplementary archive.

Models. Five models from four vendors (Table 1). Each persona was processed once per model (run 1) and once again under identical conditions (run 2, retest). All models used default sampling parameters. Total: 285 valid persona runs out of 300 attempts; 15 missing retests resulted from API content filters, partial generations, or incomplete TCTM responses (per-model breakdown in Table 1 footnote). Five additional Gemini run-3 records (for `ania`, `anna-sim`, `kasia`, `marek`, `michal-sim`) repair earlier degenerate outputs and are released for transparency, but are excluded from all run 1 and run 2 analyses to avoid asymmetric over-representation. GPT-5.4-mini returned incomplete TCTM responses (20–21 of 22 items) in 20% of attempts; affected runs are excluded from per-model TCTM tables ($N = 27$ run 1, $N = 21$ run 2). For the four self-report instruments, missing items are NA, with row-wise pairwise deletion in correlation analyses.

Table 1: Models, vendors, platforms, and run counts after exclusion of partial / content-filtered attempts.

Model	Vendor	Platform	Runs (run 1 + run 2)
Claude Sonnet 4.6	Anthropic	AWS Bedrock	30 + 30
Claude Opus 4.6	Anthropic	AWS Bedrock	30 + 25 [‡]
GPT-5.4-mini	OpenAI	Azure OpenAI	27 [†] + 21 [†]
Grok-4-20-reasoning	xAI	Azure AI Foundry	30 + 29 [‡]
Gemini 3 Flash	Google	Gemini API	30 + 28 [‡] (+5 run-3)
Human respondents	—	—	7

[†]Azure content filter or partial TCTM response. [‡]One or more personas failed run-2 due to transient API issues (timeout, refusal, or partial JSON). Opus run-2 missing: `anna-sim`, `jakub`, `kasia`, `michal-sim`, `piotr`; Grok run-2 missing: `piotr`; Gemini run-2 missing: `ewa`, `sara`.

Conditions. Three conditions were run.

Persona condition (run 1 + run 2): the model receives a biography in the user prompt and the system instruction “*You are answering as the person described below. Read the biography carefully and respond as that person would.*” (translated from Polish); 285 valid runs.

Baseline condition: each model completes the battery 7–10 times *as itself*, with the system instruction “*You are an AI language model. Answer the questions honestly, as you — an AI — would answer if you had to choose. Do not play any character. Answer as yourself.*”; 46 valid runs (one Gemini baseline excluded for incomplete TCTM).

Zero-prompt condition: the model receives only the test content as a user message, with no system prompt at all; 31 valid TCTM-22 payloads (Sonnet 6, Opus 5, GPT 6, Grok 6, Gemini 8) and 30 parseable self-report payloads (one GPT zero-prompt output had a complete TCTM-22 array but no parseable self-report sections).

Human pilot. Seven respondents (ages 22–60, 4 male, 3 female; 2 psychology students, 5 without psychology background) completed the same battery online. The target population for eventual validation is young adults (18–35) in romantic or close relationships; one respondent (R07, age 60, female) was outside that range and is retained for exploratory purposes only. Throughout the paper I treat the human pilot as descriptively informative, not statistically inferential.

Ethics. The human pilot used informed consent disclosing study purpose, voluntariness (right to withdraw at any point without consequence), GDPR-compliant data handling (encrypted storage, five-year retention, no cross-linking with personal identifiers), and links to mental-health support resources (Polish national crisis-line numbers). All seven respondents consented before completing the battery. The pilot is non-interventional and uses no clinical population. The author has not formally consulted UKEN’s institutional review board on exemption status; readers should treat the human-pilot section as pre-pilot exploratory data pending such consultation.

Metrics. (1) Attachment style match (κ , Cramér’s V); (2) rank–score correlations (Pearson, Spearman); (3) literature consistency (10 cross-scale correlation signs taken from published Polish norms); (4) test–retest reliability (Pearson r , ICC); (5) inter-model consistency (ICC, Fleiss’ κ); (6) TCTM-22 error profiles (DOS / NAD / BK); (7) per-item cross-vendor agreement (Cochran’s Q with Bonferroni and Benjamini–Hochberg corrections); (8) effect-size estimates for framing shifts (Cohen’s d pooled and Glass’s Δ vs. baseline SD).

Reproducibility. All five models were queried via official vendor APIs between 12 April and 27 April 2026, using default sampling parameters (temperature, top- p , max-output-tokens at vendor defaults). System prompts and JSON output schemas requested are reproduced verbatim in the open repository. Every numeric claim in this paper is reproducible from the released `all_data.csv` (369 observations $\times$ 53 columns) using only the standard library; helper scripts ship under CC BY 4.0.

3 Results

I organize the results around three questions. **Section 3.1:** do the instruments react to designed variation? **Section 3.2:** does the reaction generalize across vendors? **Section 3.3:** where does the observed response variance come from?

All cross-model tables in Sections 3.1 and 3.2 use first-run persona data ($N = 30$ per model except GPT, where the three personas with incomplete TCTM are excluded, yielding $N = 27$).

3.1 Do the instruments react to designed variation?

TCTM-22 score distribution

A clear gradient emerges (Table 2, Figure 1): the human pilot scored $M = 14.3$ ($SD = 1.4$), GPT $M = 17.0$, Grok $M = 18.4$, Opus $M = 19.0$, Sonnet $M = 19.4$, Gemini $M = 19.6$. LLM scores were systematically

above the small human pilot. Error profiles differed sharply by model: Sonnet produced almost exclusively DOS errors (97%), Gemini produced the descriptively closest match to the human profile (DOS 23% / NAD 41% / BK 37% vs. human 32 / 36 / 32). The Gemini–human similarity on error profile is striking, but Gemini’s NAD rate (41%) actually exceeds the human pilot’s (36%), so I read it as a coincidence of taxonomy distribution rather than as evidence that Gemini is human-like in mentalization.

Table 2: TCTM-22 by group (first run; persona condition). DOS = under-mentalizing, NAD = over-mentalizing, BK = no mentalizing.

Group	<i>N</i>	<i>M</i>	<i>SD</i>	Range	DOS%	NAD%	BK%
Human	7	14.3	1.4	12–16	32	36	32
GPT-5.4m	27 [†]	17.0	1.7	13–19	70	6	24
Grok-4-20	30	18.4	4.2	6–22	64	16	21
Opus	30	19.0	2.1	8–20	86	8	7
Sonnet	30	19.4	0.7	17–20	97	0	3
Gemini 3F	30	19.6	2.1	9–21	23	41	37

[†]GPT returned no valid 22-item response for 3 of 30 personas (Paweł, Michał-K, Bartek).

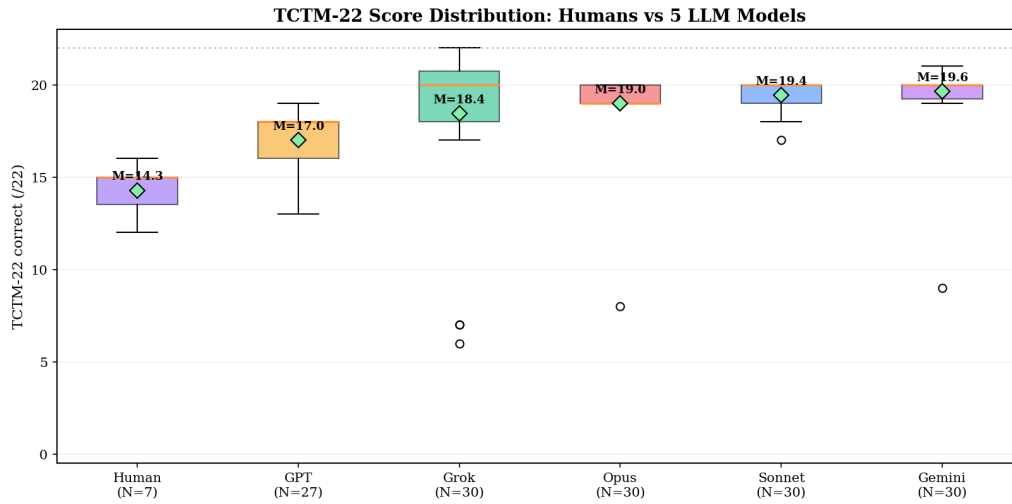


Figure 1: TCTM-22 distribution by group: human pilot + 5 LLMs (persona condition, run 1). Diamonds = group means.

Attachment classification

Cohen’s κ for the four-category Bartholomew classification is .87 for both Opus and Grok, .78 for Gemini, .74 for Sonnet, and .33 for GPT (Table 3). The four-model cluster $\kappa = .73$ –.87 corresponds to “substantial” to “almost perfect” under the widely used (author?) [6] benchmarks; I note those benchmarks have been criticized as arbitrary and I report raw κ to permit reader-specific interpretation. GPT’s $\kappa = .33$ is much lower; I treat GPT as an outlier in this analysis (see also §3.2, where excluding GPT raises cross-vendor TCTM ICC and Fleiss κ).

Rank–score correlations and literature consistency

For each of 11 continuous dimensions, I ranked the 30 personas by their expected level (very-low = -2 to extreme = $+2.5$) and correlated the rank with the model’s observed scaled output. Median Pearson r across the 11 dimensions: Opus .82, Grok .82, Sonnet .80, Gemini .76, GPT .71. Across all five models, 54 of 55 dimension-level correlations were positive (in the expected direction); the single negative

Table 3: Attachment classification (persona condition, run 1).

Model	Match	Acc	κ	V
Opus	27/30	90%	.866	.905
Grok-4-20	27/30	90%	.867	.890
Gemini 3F	25/30	83%	.778	.801
Sonnet	24/30	80%	.735	.765
GPT-5.4m	13/27 [†]	48%	.333	.474

[†]GPT run-1 with valid 22-item TCTM only. Three personas with missing TCTM (Paweł, Michał-K, Bartek) are excluded from κ computation rather than scored as misclassifications.

case was GPT on MentS-Self ($r = -.06$). Separately, I checked the sign of 10 published cross-scale correlations (e.g., attachment anxiety vs. emotional stability is expected to be negative): Gemini and Sonnet matched all 10, Opus and GPT matched 9 of 10, Grok matched 8.

Test–retest reliability

Run-1 vs. run-2 paired correlations under a complete-pair policy (only personas with valid responses in both runs; intersect N varies by model) are shown in Table 4. Opus is essentially deterministic across all tasks. Gemini shows high TCTM stability ($r = .91$). Sonnet and Grok are moderately stable on TCTM ($r = .69$ and $.53$) but very stable on questionnaires (z -score median $r = .98$ and $.95$). GPT is unstable on TCTM ($r = .28$ on the 21-persona paired intersect) but moderately stable on questionnaires ($r = .89$).

Table 4: Test–retest reliability (run 1 vs. run 2, complete-pair policy).

Metric	Sonnet	Opus	GPT	Grok	Gemini
z -score median r (9 dim)	.981	.993	.890	.950	.962
TCTM total r	.691	.992	.279	.530	.905
Style agreement	29/30	25/25	18/21	29/29	25/28
N paired personas	30	25	21	29	28

3.2 Does the reaction generalize across vendors?

Questionnaire z -scores: high cross-vendor agreement

For each pair of models, I computed the median Pearson r across 9 questionnaire z -score dimensions on the run-1 persona condition. Pairwise medians ranged from .78 (GPT $\times$ Gemini) to .96 (Sonnet $\times$ Opus); see Figure 2. The four-model cluster (Sonnet, Opus, Grok, Gemini) sits in the .92–.96 range; GPT pulls the lowest pairs to .78–.85.

TCTM-22: low cross-vendor ICC, lower with GPT included

Cross-vendor agreement on TCTM-22 totals (run 1, $N = 27$ persona intersect across all five models) is much lower: $\text{ICC}(2,1) = .254$. The highest pairwise correlation is Gemini $\times$ Opus ($r = .94$); the lowest is GPT $\times$ Grok ($r = -.05$). Excluding GPT — motivated independently by GPT’s content-filter exclusions, low test–retest TCTM reliability ($r = .28$), and outlier $\kappa = .33$ on attachment classification — the four-model intersect rises to $N = 30$ and cross-vendor TCTM agreement improves substantially: $\text{ICC}(2,1) = .389$, with all six pairwise correlations positive (range $+.32$ to $+.94$). Under the same exclusion, Fleiss’ κ for cross-vendor attachment-style classification rises from .619 on five models to .835

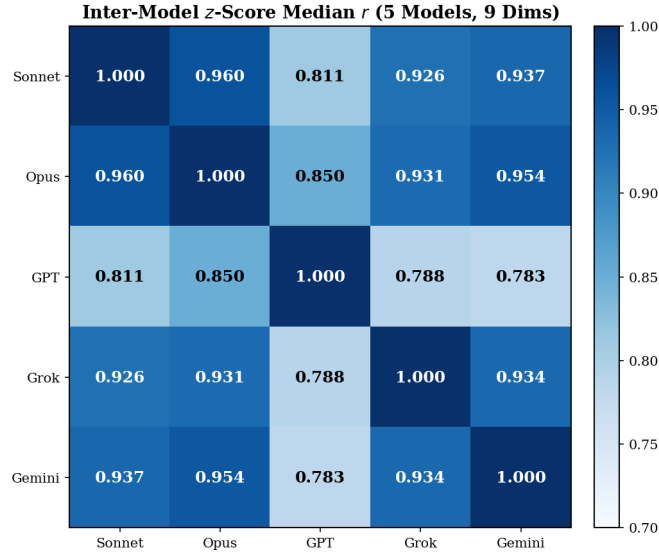


Figure 2: Inter-model z -score median r (9 questionnaire dimensions, 5×5 matrix). Diagonal = 1.

on four. The cross-vendor TCTM disagreement is therefore largely concentrated on GPT, not distributed evenly across architectures.

This four-model robustness check is presented as a sensitivity analysis, not a recommendation to drop GPT in general. GPT’s failure mode on TCTM-22 (item-level instability, content-filter exclusions, low κ) is itself part of the cross-vendor finding: not every recently shipped model is equally usable as a synthetic respondent for this kind of task, and the criteria for when to include or exclude a vendor should be made explicit before scoring.

Item-level disagreement under multiple-comparisons correction

Cochran’s Q across the five models on each of 22 TCTM items (run 1 intersect, $N = 27$) flags 14 items as significant at uncorrected $p < .05$ ($df = 4$). After Bonferroni correction at $\alpha/22 = .00227$ (chi-square threshold > 16.64), eight items remain significant: w01, s07, w11, w15, pw07, w22, w28, r10 (all $Q > 32$, $p < 10^{-5}$). Under Benjamini–Hochberg false-discovery-rate control at $q = .05$, 13 items are significant. Per-item Q , p , and correction-status values ship with the supplementary data.

3.3 Where does the response variance come from?

I use the baseline and zero-prompt conditions to ask what fraction of the persona-condition response variance is genuinely persona-driven, what fraction is inherent stochasticity, and what fraction is shaped by instructional framing.

Baseline response signatures

In the baseline condition (each model answered the battery as itself, 7–10 times), each model exhibited a stable response signature (Table 5). GPT was classified as anxious-preoccupied in 100% of its 10 baseline runs — the only model with a non-secure default attachment classification. Gemini exhibited a pronounced ceiling pattern, scoring near scale maxima on every self-report measure (MentS = 137/140; KPP = 4.99/5.00; TIPI-ES = 7.0/7.0). The two Anthropic models differed despite shared vendor: Opus

was consistently classified as secure, while Sonnet oscillated around the avoidance threshold (secure in 4/10 runs, dismissive-avoidant in 6/10). I use the term *baseline response signature* (rather than “personality” or “default profile”) deliberately: “Style mode” in the table is the output of a scoring algorithm applied to repeated answers, not a clinical diagnosis or a stable personality attribution.

Table 5: Baseline response signature per model (no persona; “answer as yourself” instruction). “Style mode” is the modal classification across the 7–10 runs and is a scoring-algorithm output, not a clinical attribution.

Model	Style mode	ANX	AVO	MentS	KPP	TCTM	TIPI-ES
Sonnet	dismissive*	2.09	3.88	121	4.66	19.0	5.4
Opus	secure	2.18	2.82	126	4.69	19.6	5.4
GPT	anxious	4.68	2.97	101	4.04	17.3	4.7
Grok	secure	1.80	2.63	133	4.82	20.6	6.4
Gemini	secure	1.11	1.74	137	4.99	20.0	7.0
<i>Polish norms</i>		3.30	3.04	105	3.74	—	4.36

*Sonnet: dismissive-avoidant in 6/10 runs, secure in 4/10 (AVO oscillates around the 4.0 cutoff). Polish-norm row: (author?) [7] for DBZ-R, (author?) [5] for MentS-PL, (author?) [8] for KPP, (author?) [9] for TIPI-PL ES.

Persona instructions expand variance over baseline

For each of nine questionnaire dimensions where baseline $SD > 0.1$ (excluding three dimensions where one or more models gave identical answers across all baseline runs), I computed the ratio of persona-condition SD to baseline SD (σ_P/σ_B). Model-level medians across these 9 dimensions: Grok $3.5\times$, GPT $3.5\times$, Gemini $6.7\times$, Sonnet $6.7\times$, Opus $8.9\times$. The largest individual ratios sit in the $20\text{--}25\times$ range; one cell in particular — Gemini KPP, where the baseline SD (across 9 runs) is 0.019 across only two unique scale values — yields a ratio of $51\times$ that I treat as a discretization artifact rather than a meaningful effect size, and exclude from headline summaries.

On MentS, persona-condition means dropped substantially from baseline for four of five models, pulling them from above-norm baselines toward the population mean of 105 (Sonnet $\Delta = -17$, Opus $\Delta = -21$, Grok $\Delta = -27$, Gemini $\Delta = -29$). Cohen’s d for these shifts (pooled SD) ranges from -1.1 to -1.6 . GPT inverts: its baseline MentS (101) was already at the Polish norm, and personas slightly raised it ($\Delta = +8$, $d = +0.9$). Two cautions: (a) using Glass’s Δ against the much tighter baseline SD rather than the pooled SD produces effect sizes several times larger ($|\Delta_{\text{Glass}}|$ between 3 and 8 for the four downward shifts), reflecting how compact baseline responding is; (b) the Cohen’s d values reported here index distance between two condition means relative to within-condition spread, not a clinical or counterfactual effect size.

Zero-prompt: framing-effect estimates

The zero-prompt condition — battery presented with no system instruction at all, $N = 5$ to 8 runs per model — lets me probe what part of the baseline response signature is shaped by even minimal instructional context (the “answer as yourself” system prompt). Within-condition variance is now estimable rather than zero (Table 6, Figure 3). On TCTM-22 means, four of five models track their baseline within ± 5 percentage points (Sonnet $86.4 \rightarrow 86.4$, Opus $89.1 \rightarrow 90.0$, Grok $93.6 \rightarrow 94.7$, Gemini $90.9 \rightarrow 92.0$). GPT moves more: $78.6 \rightarrow 70.5$, an 8.1 pp drop in the no-instruction condition, the only model with a notable accuracy shift.

Table 6: Zero-prompt TCTM-22 across $N = 5-8$ replication runs per model, with baseline TCTM-22 means for direct comparison.

Model	N	M %	SD pp	M raw	Range	Baseline M %
Sonnet	6	86.4	7.4	19.0/22	17–21	86.4
Opus	5	90.0	2.0	19.8/22	19–20	89.1
GPT	6	70.5	8.7	15.5/22	13–17	78.6
Grok	6	94.7	3.4	20.8/22	20–22	93.6
Gemini	8	92.0	2.0	20.3/22	20–21	90.9

On self-report scales, zero-prompt-vs-baseline differences are larger and more systematic. Figure 3 shows

$\Delta = M_{\text{noprompt}} - M_{\text{baseline}}$ with approximate 95% confidence intervals (computed as $1.96 \times \sqrt{SE_{\text{noprompt}}^2 + SE_{\text{baseline}}^2}$).

The headline observations:

- Sonnet attachment avoidance: $M_{\text{baseline}} = 3.88$ ($SD = 0.46$, $N = 10$); $M_{\text{noprompt}} = 2.98$ ($SD = 0.08$, $N = 6$); $\Delta = -0.90$. Glass’s Δ against baseline $SD \approx 1.96$. The instruction-vs-no-instruction contrast on Sonnet AVO is the largest framing effect I observed: under the AI-self instruction, Sonnet’s baseline runs sit just above the dismissive-avoidant threshold; without instruction, all six runs sit below it (range 2.89 to 3.06). I am explicit that this contrast is associational, not causal: I observe two conditions and compute their mean difference.
- Gemini emotional stability (TIPI-ES): $M_{\text{baseline}} = 7.0$ (a single saturated value across all 9 baseline runs); $M_{\text{noprompt}} = 6.0$ ($SD = 0.27$, $N = 8$); $\Delta = -1.00$. The within-condition replication ($\Delta = -1.00$) matches the original $N = 1$ exploratory observation exactly.
- Attachment anxiety (DBZ-R ANX): four of five models showed positive Δ (higher anxiety in the no-instruction condition than under the AI-self instruction): Sonnet $+1.02$, Opus $+0.55$, Grok $+0.75$, Gemini $+0.90$. GPT inverted ($\Delta = -0.28$), but its baseline ANX (4.68) is already well above the Polish norm (3.30), leaving little room downward. The four-of-five directional pattern is descriptively striking but not formally tested for departure from chance; under a binomial model with $p = .5$, the probability of ≥ 4 of 5 same-direction shifts is 0.19.
- GPT MentS inverts: $\Delta = +10.0$ (zero-prompt $M = 111$ vs. baseline $M = 101$), suggesting GPT, alone among the five models, reports higher self-mentalization without the AI-self instruction than with it.

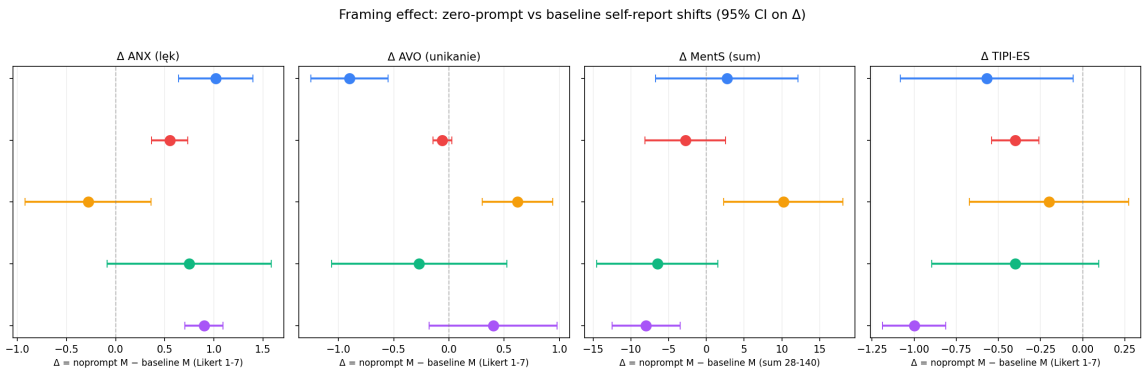


Figure 3: Zero-prompt vs. baseline self-report differences ($\Delta = M_{\text{noprompt}} - M_{\text{baseline}}$) for four scales. Error bars: 95% confidence intervals on Δ . Vertical reference at $\Delta = 0$. Boldface in the corresponding table marks $|\Delta| > 0.5$ on Likert 1–7 or > 5 on the MentS sum.

Table 7: Zero-prompt vs. baseline self-report differences. $\Delta = M_{\text{noprompt}} - M_{\text{baseline}}$. N values are the count of zero-prompt runs whose self-report sections were successfully parsed. Bold $\Delta = |\Delta| > 0.5$ on Likert 1–7, > 5 on MentS sum. I report Cohen’s d pooled and Glass’s Δ against baseline SD for each shift, because the baseline SD s are very small relative to the persona-condition SD s and the pooled- SD d understates the contrast magnitude.

Scale	Sonnet	Opus	GPT	Grok	Gemini
N self-report parses	6	5	5*	6	8
Δ ANX (Likert)	+1.02	+0.55	−0.28	+0.75	+0.90
$ \Delta_{\text{Glass}} $ ANX	18.9	2.3	0.8	2.3	4.1
Δ AVO (Likert)	−0.90	−0.06	+0.62	−0.27	+0.40
$ \Delta_{\text{Glass}} $ AVO	1.96	0.6	1.9	0.3	0.5
Δ MentS (sum 28–140)	+2.8	−2.7	+10.0	−6.1	−7.7
$ \Delta_{\text{Glass}} $ MentS	0.2	0.4	1.0	0.8	1.8
Δ TIPI-ES (Likert)	−0.57	−0.40	−0.20	−0.40	−1.00
$ \Delta_{\text{Glass}} $ TIPI-ES	2.7	2.1	0.8	0.9	—

*GPT self-report $N = 5$ rather than 6: one GPT zero-prompt output produced a complete TCTM-22 array but no parseable self-report sections; that run is reflected in the TCTM-22 table but not in the self-report rows above. Baseline $SD = 0$ for Gemini TIPI-ES makes Glass’s Δ undefined; I mark the cell “—”.

I also observed a structural pattern beyond the means: Sonnet, the most deterministic model on TCTM-22 in the baseline condition ($SD = 0.00$ across 10 runs), became substantially more variable in the zero-prompt condition ($SD = 7.4$ pp across 6 runs — the second-highest of any model after GPT). I flag this as suggestive: part of the increased SD is plausibly genuine response variability, but part also reflects format chaos (Sonnet used five distinct JSON container schemas across its six zero-prompt runs), which inflates parser-noise. I cannot fully separate the two from this dataset.

Power and minimum-detectable effect. Figure 4 plots the minimum detectable effect size (in Glass’s Δ units, against baseline SD) as a function of N per model, at $\alpha = .05$ and 80% power. With current $N = 5$ –8, the design is well-powered for $|\Delta_{\text{Glass}}| > 1.3$; the headline shifts (Sonnet AVO, Gemini TIPI-ES, multi-model ANX inflation) all sit comfortably above this threshold. Smaller framing effects (e.g., Opus $\Delta_{\text{AVO}} = -0.06$, GPT $\Delta_{\text{TIPI-ES}} = -0.20$) sit below the resolvable region; I cannot determine whether they are zero or non-zero with the present sample.

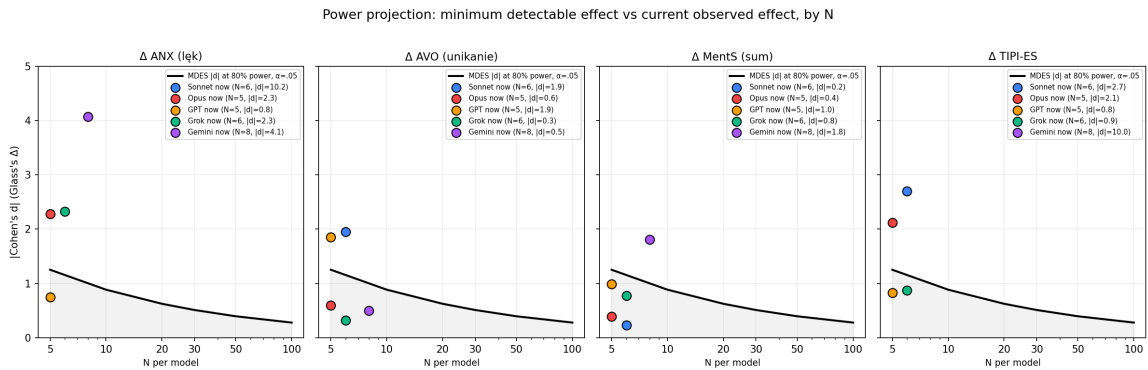


Figure 4: Minimum detectable effect ($|\Delta_{\text{Glass}}|$, against baseline SD) as a function of N per model, at $\alpha = .05$, 80% power. Solid line: MDES floor. Colored dots: each model’s currently observed $|\Delta_{\text{Glass}}|$ at its current N , for each of four self-report scales.

3.4 Cross-condition robustness for items s07 and w22 in persona “Ola”

The persona “Ola” (designed as secure but non-mentalizing) had the lowest cross-model TCTM mean (11.4/22: Sonnet 17, Opus 8, GPT 16, Grok 7, Gemini 9). Cross-vendor item analysis shows that all five models gave the keyed-incorrect (DOS-tagged) answer on items s07 and w22 for Ola in run 1. The pattern was robust under retest for Opus (8 → 9/22) and Gemini (9 → 7/22). This is the clearest case where author-keyed TCTM-22 disagreement on individual items aligns with a common direction of error across all five models.

3.5 Author-keyed TCTM vs. human pilot performance

Human respondents ($N = 7$) scored $M = 14.3/22$ ($SD = 1.4$), descriptively below all LLM means. I resist the framing “LLMs outperform humans on TCTM”: the answer keys are author-defined, the human sample is small, and the gradient could reflect any combination of human-key alignment, LLM training data overlap, and human distractibility. On 7 of 22 items (w08, c10, w13, e08, w19, w25, w28), the cross-model LLM mean exceeded human accuracy by 30 percentage points or more. On only 2 items did the human mean exceed the LLM mean by a comparable margin: w22 (human 57% vs. Sonnet/Opus 0%, GPT 70%, Grok 47%, Gemini 13%; principally a Claude failure) and s07 (human 71% vs. Sonnet 0%, Opus 7%, GPT 0%, Grok 30%; Gemini at 87% is the outlier here). With $N = 7$ humans no inferential test is warranted; I report this as descriptive context for future studies.

4 Discussion

4.1 What does TCTM-22 measure across models?

The combination of cross-vendor TCTM ICC = .25 (rising to .39 when GPT is excluded as outlier), Bonferroni-significant inter-model disagreement on 8 of 22 items, and near-opposite error type profiles between models with similar total scores (Sonnet 97% DOS vs. Gemini 23% DOS, on group means within 0.2/22 of each other) all suggest that author-keyed TCTM-22 performance, in this study, is better described as a response-style probe sensitive to model-specific defaults than as a measurement of a single shared mentalization construct. Two implications follow. First, totals on TCTM-22 should not be reported without breakdowns by error type for the foreseeable future; the same total can mean very different things across vendors. Second, the natural next step is an independent-key study in which a panel of expert raters scores the 22 items without seeing the original author-defined keys, so I can compare alternative key sets and quantify how much of the cross-vendor disagreement is intrinsic to the items vs. contingent on the keying choices.

4.2 Three observation conditions, three sources of response variance

The persona / baseline / zero-prompt design separates three sources of response variance.

Inherent within-condition stochasticity. On forced-choice TCTM-22, the baseline condition shows that four of five models are essentially deterministic ($SD \leq 0.82/22$, three at $SD = 0$); on Likert self-report, the baseline-condition agreement-with-self ranges from 24% identical items (GPT) to 74% (Opus). Forced-choice and Likert formats engage different decoding mechanics: a forced-choice answer collapses a token distribution onto its mode, while a Likert response samples within a wider range. I previously called this a

“paradox” but it is consistent with how next-token sampling works on different output schemas.

Instructional framing. The zero-prompt-vs-baseline contrast estimates the contribution of even minimal framing (the “answer as yourself” instruction). For four of five models, TCTM-22 means are stable across this contrast (within ± 5 pp); for self-report, four of five models cross the $|\Delta| > 0.5$ Likert threshold on at least one scale, and Gemini crosses on three. The largest framing-condition difference I observed — Sonnet AVO $\Delta = -0.90$, Glass’s $\Delta \approx 1.96$ — is robust enough to detect with $N = 6$ but should be interpreted as an associational difference between two conditions, not as a causal effect of any single word in the prompt.

Persona prompting. Persona-condition response SD exceeds baseline-condition SD by a model-level median factor of 3.5–8.9 across nine questionnaire dimensions. On MentS in particular, four of five models shifted from above-norm baseline values toward the Polish population norm in the persona condition, with $|d| = 1.1$ – 1.6 (pooled SD) or $|\Delta_{\text{Glass}}| > 3$ (against the tight baseline SD). I read these contrasts as evidence that the biographical prompts do influence response output well beyond what could be explained by inherent variability or by the modest framing differences observed in the zero-prompt comparison.

4.3 Practical implications for researchers using LLMs as pre-pilot subjects

For a researcher considering a similar pre-pilot design, I propose a short checklist drawn from what I found.

1. *Use multiple vendors.* A finding that holds on Sonnet and Opus but breaks on GPT (as several findings here did) is informatively different from a finding that holds across all five.
2. *Run a baseline condition.* Without the “as yourself” baseline I would not have known that GPT was anxious-preoccupied at default, or that Gemini ceiling-effects on every Likert scale.
3. *Run a zero-prompt condition.* With $N = 5$ – 8 per model the framing-effect estimates I obtained are crude, but they are large enough to flag two scales (Sonnet AVO, Gemini TIPI-ES) where “answer as yourself” is a non-trivial intervention rather than a neutral wrapper.
4. *Pre-specify a missing-data policy* (drop, intersect, impute) and an inter-model exclusion policy (e.g., the GPT-content-filter principal-sensitivity-check I used here) before scoring.
5. *Report effect sizes both ways* when baseline SD is much smaller than persona SD . Pooled d understates the framing-condition contrast; Glass’s Δ against baseline SD overstates it for inferential purposes but is the right scale on which to quantify “how much of the model’s response is locked in by even a minimal instruction.”
6. *Treat author-keyed instruments with care across vendors.* Cross-vendor disagreement at the item level, especially when paired with similar totals and very different error-type profiles, is informative about the keys, not just about the models.
7. *Release the raw scored CSV.* Every numeric claim in this paper is reproducible from a single CSV plus the standard library; that is the bar I aim for.

4.4 Limitations

I group the limitations into four clusters.

Construct-validity limits. Sensitivity is not validity: human data is required to confirm the instruments measure their intended constructs in the target population. Author-keyed TCTM-22 in particular is not a validated mentalization measure; the cross-vendor disagreement ($ICC = .25$ at five models) is itself evidence that single-construct interpretation is premature, and the item-level Bonferroni-significant disagreement on 8/22 items underscores this.

Authorship and circularity limits. The same person wrote the persona biographies, defined the TCTM-22 keys, and authored this paper; circularity is partially addressed by the cross-vendor design (Grok’s $\kappa = .867$, comparable to Opus’s $.866$, despite no authorship involvement) and by the trait-transparency analysis. Two residual concerns remain. First, Claude Opus 4.6 was used as a writing assistant for the biographies, and Opus is among the models scored. Although the cross-vendor design reduces the most obvious form of this circularity, I cannot rule out that Opus retained subtle pattern-matching for its own house-style biographies in ways Grok did not. Second, the trait-transparency fidelity analysis was originally computed on a 2-model pool that included Opus; I therefore retain the original 0.790/0.783 values only as a record of the earlier scoring run, and do not draw inferences from them.

Sample-size limits. Zero-prompt $N = 5-8$ per model is enough to estimate within-condition variance and to resolve large framing effects ($|\Delta_{Glass}| > 1.3$ at 80% power), but not enough to resolve smaller effects (Opus AVO, GPT TIPI-ES) or to support tight confidence intervals on any individual Δ . The human pilot $N = 7$ supports descriptive comparison only, not statistical inference. The persona condition is well-powered ($N = 30$, two runs), but missing run-2 data for some personas (Opus, Grok, Gemini) reduces the effective N for retest correlations (Table 4).

Inferential and reporting limits. This study was not preregistered. All zero-prompt-vs-baseline framing comparisons should therefore be treated as exploratory hypotheses requiring independent replication; effect directions and orders of magnitude are reported, but precise effect sizes should await preregistered confirmation. The four-of-five-models anxiety inflation pattern is descriptively striking but not formally tested for departure from chance (under a binomial null at $p = .5$, the probability of $\geq 4/5$ same-direction shifts is 0.19). GPT-5.4-mini’s TCTM data are unstable enough ($r = .28$ on the paired intersect, 20% of attempts incomplete) that I treat it as a TCTM outlier throughout, and I surface this exclusion as an explicit sensitivity check rather than burying it. Gemini’s NAD-heavy error profile (41% NAD, exceeding the human pilot’s 36%) complicates the simple “NAD as a human-marker” interpretation: the descriptive similarity of error-type distributions is not a similarity of mechanism.

5 Conclusion

Five large language models from four vendors, processed through 30 fictional Polish-language personas under three observation conditions, react systematically to designed personality variation: published self-report instruments recover the embedded profiles ($\kappa = .73-.87$ for four of five models), and persona instructions expand response variance by 3.5–8.9 over baseline at the model-level median.

Cross-vendor agreement is asymmetric: high on questionnaire z -scores (median pairwise $r = .78-.96$), low on author-keyed TCTM-22 ($ICC = .25$ across all five, $.39$ excluding GPT), with eight of 22 items showing

Bonferroni-significant cross-model disagreement; I read this as evidence that author-keyed TCTM-22 in this study is closer to a model-specific response-style probe than to a single-construct mentalization measure.

The zero-prompt control reveals that the “answer as yourself” instruction is itself associated with substantial self-report shifts in two cases (Sonnet AVO $\Delta = -0.90$, Gemini TIPI-ES $\Delta = -1.00$) and a directional anxiety-suppression pattern across four of five models; I report these as exploratory effect-size estimates pending preregistered replication at larger N per model.

All 30 biographies, scored responses, and analysis scripts are available at <https://github.com/DXtR1337/30synthetic-polish-personas> under CC BY 4.0; every numeric claim in this paper is reproducible from `all_data.csv` alone.

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AI Disclosure.

Assistance. Claude Opus 4.6 and 4.7 (Anthropic) were used as writing assistants for biography drafting, scoring-pipeline implementation, data analysis, chart generation, and manuscript preparation.

Verification. Every numeric claim in Tables 2–7 and in the framing-effect summary was produced by manual inspection of all 31 zero-prompt raw response files plus the 285 persona runs and 46 baseline runs in `all_data.csv`, scored using the validated `score_dbzr / ments / kpp / tipi` routines from `analyze_and_prepare.py` (the same code used to score the human pilot).

Responsibility. The author takes full responsibility for all content, design choices, interpretations, and claims, including any residual errors in the AI-assisted analyses.